

MySQL Professional Student Assessment

Part 1: Database Design

1 Create a database named student_management

```
1 CREATE DATABASE student_management;  
2 USE student_management;
```

2 Create tables: Students, Courses, Enrollments, Marks

```
1 CREATE TABLE Students (  
2     student_id INT PRIMARY KEY,  
3     name VARCHAR(50) NOT NULL,  
4     email VARCHAR(100) UNIQUE,  
5     phone VARCHAR(15),  
6     department VARCHAR(50) DEFAULT 'CSE'  
7 );  
8  
9 CREATE TABLE Courses (  
10     course_id INT PRIMARY KEY,  
11     course_name VARCHAR(50) NOT NULL,  
12     credits INT  
13 );  
14  
15 CREATE TABLE Enrollments (  
16     enrollment_id INT PRIMARY KEY,  
17     student_id INT,  
18     course_id INT,  
19     FOREIGN KEY (student_id) REFERENCES Students(student_id),  
20     FOREIGN KEY (course_id) REFERENCES Courses(course_id)  
21 );  
22  
23 CREATE TABLE Marks (  
24     mark_id INT PRIMARY KEY,  
25     student_id INT,  
26     course_id INT,  
27     marks INT,  
28     grade CHAR(2),  
29     FOREIGN KEY (student_id) REFERENCES Students(student_id),  
30     FOREIGN KEY (course_id) REFERENCES Courses(course_id)  
31 );
```

Part 2: Data Insertion

1. Insert sample data: minimum 10 students, 5 courses, 15 enrollments, and 20 marks.

Run SQL query/queries on database student_management:

```
1 INSERT INTO Students VALUES  
2 (1, 'Arun', 'arun@gmail.com', '9876543210', 'CSE'),  
3 (2, 'Bala', 'bala@gmail.com', '9876543211', 'ISE'),  
4 (3, 'Charan', 'charan@gmail.com', '9876543212', 'ECE'),  
5 (4, 'Divya', 'divya@gmail.com', '9876543213', 'CSE'),  
6 (5, 'Esha', 'esha@gmail.com', '9876543214', 'ISE'),  
7 (6, 'Farhan', 'farhan@gmail.com', '9876543215', 'ME'),  
8 (7, 'Goku1', 'goku1@gmail.com', '9876543216', 'CSE'),  
9 (8, 'Hari', 'hari@gmail.com', '9876543217', 'ECE'),  
10 (9, 'Isha', 'isha@gmail.com', '9876543218', 'ISE'),  
11 (10, 'John', 'john@gmail.com', '9876543219', 'CSE');
```

Run SQL query/queries on database student_manage

```
1 INSERT INTO Courses VALUES
2 (101, 'DBMS', 4),
3 (102, 'Java', 3),
4 (103, 'Python', 3),
5 (104, 'Maths', 4),
6 (105, 'OS', 3);
```

```
1 INSERT INTO Marks VALUES
2 (1, 1, 101, 85, 'A'),
3 (2, 1, 102, 78, 'B'),
4 (3, 2, 101, 88, 'A'),
5 (4, 3, 103, 67, 'C'),
6 (5, 4, 104, 90, 'A'),
7 (6, 5, 105, 72, 'B'),
8 (7, 6, 101, 60, 'C'),
9 (8, 7, 102, 95, 'A'),
10 (9, 8, 103, 55, 'C'),
11 (10, 9, 104, 80, 'B'),
12 (11, 10, 105, 70, 'B'),
13 (12, 2, 103, 65, 'C'),
14 (13, 3, 104, 75, 'B'),
15 (14, 4, 105, 82, 'A'),
16 (15, 5, 101, 77, 'B'),
17 (16, 6, 102, 69, 'C'),
18 (17, 7, 103, 91, 'A'),
19 (18, 8, 104, 73, 'B'),
20 (19, 9, 105, 68, 'C'),
21 (20, 10, 101, 85, 'A');
```

Run SQL query/queries on database student_management

```
1 INSERT INTO Enrollments VALUES
2 (1, 1, 101),
3 (2, 1, 102),
4 (3, 2, 101),
5 (4, 3, 103),
6 (5, 4, 104),
7 (6, 5, 105),
8 (7, 6, 101),
9 (8, 7, 102),
10 (9, 8, 103),
11 (10, 9, 104),
12 (11, 10, 105),
13 (12, 2, 103),
14 (13, 3, 104),
15 (14, 4, 105),
16 (15, 5, 101);
```

Part 3: SQL Queries

1 Basic queries: SELECT, WHERE, ORDER BY, LIMIT

a). SELECT * FROM Students

		student_id	name	email	phone	department
☐	Edit Copy Delete	1	Arun	arun@gmail.com	9876543210	CSE
☐	Edit Copy Delete	2	Bala	bala@gmail.com	9876543211	ISE
☐	Edit Copy Delete	3	Charan	charan@gmail.com	9876543212	ECE
☐	Edit Copy Delete	4	Divya	divya@gmail.com	9876543213	CSE
☐	Edit Copy Delete	5	Esha	esha@gmail.com	9876543214	ISE
☐	Edit Copy Delete	6	Farhan	farhan@gmail.com	9876543215	ME
☐	Edit Copy Delete	7	Gokul	gokul@gmail.com	9876543216	CSE
☐	Edit Copy Delete	8	Hari	hari@gmail.com	9876543217	ECE
☐	Edit Copy Delete	9	Isha	isha@gmail.com	9876543218	ISE
☐	Edit Copy Delete	10	John	john@gmail.com	9876543219	CSE

☐ Check all With selected: Edit Copy Delete Export

b). SELECT * FROM Students WHERE department='CSE';

		student_id	name	email	phone	department
☐	Edit Copy Delete	1	Arun	arun@gmail.com	9876543210	CSE
☐	Edit Copy Delete	4	Divya	divya@gmail.com	9876543213	CSE
☐	Edit Copy Delete	7	Gokul	gokul@gmail.com	9876543216	CSE
☐	Edit Copy Delete	10	John	john@gmail.com	9876543219	CSE

☐ Check all With selected: Edit Copy Delete Export

c). SELECT * FROM Students ORDER BY name;

		student_id	name	email	phone	department
☐	Edit Copy Delete	1	Arun	arun@gmail.com	9876543210	CSE
☐	Edit Copy Delete	2	Bala	bala@gmail.com	9876543211	ISE
☐	Edit Copy Delete	3	Charan	charan@gmail.com	9876543212	ECE
☐	Edit Copy Delete	4	Divya	divya@gmail.com	9876543213	CSE
☐	Edit Copy Delete	5	Esha	esha@gmail.com	9876543214	ISE
☐	Edit Copy Delete	6	Farhan	farhan@gmail.com	9876543215	ME
☐	Edit Copy Delete	7	Gokul	gokul@gmail.com	9876543216	CSE
☐	Edit Copy Delete	8	Hari	hari@gmail.com	9876543217	ECE
☐	Edit Copy Delete	9	Isha	isha@gmail.com	9876543218	ISE
☐	Edit Copy Delete	10	John	john@gmail.com	9876543219	CSE

☐ Check all With selected: Edit Copy Delete Export

d). SELECT * FROM Students LIMIT 5;

		student_id	name	email	phone	department
☐	Edit Copy Delete	1	Arun	arun@gmail.com	9876543210	CSE
☐	Edit Copy Delete	2	Bala	bala@gmail.com	9876543211	ISE
☐	Edit Copy Delete	3	Charan	charan@gmail.com	9876543212	ECE
☐	Edit Copy Delete	4	Divya	divya@gmail.com	9876543213	CSE
☐	Edit Copy Delete	5	Esha	esha@gmail.com	9876543214	ISE

☐ Check all With selected: Edit Copy Delete Export

2. Logical operators: AND, OR, NOT

a). `SELECT * FROM Students WHERE department='CSE' AND name LIKE 'A%';`

	student_id	name	email	phone	department
☐ Edit Copy Delete	1	Arun	arun@gmail.com	9876543210	CSE

↑ ☐ Check all With selected: Edit Copy Delete Export

b). `SELECT * FROM Students WHERE department='ISE' OR department='ECE';`

	student_id	name	email	phone	department
☐ Edit Copy Delete	2	Bala	bala@gmail.com	9876543211	ISE
☐ Edit Copy Delete	3	Charan	charan@gmail.com	9876543212	ECE
☐ Edit Copy Delete	5	Esha	esha@gmail.com	9876543214	ISE
☐ Edit Copy Delete	8	Hari	hari@gmail.com	9876543217	ECE
☐ Edit Copy Delete	9	Isha	isha@gmail.com	9876543218	ISE

↑ ☐ Check all With selected: Edit Copy Delete Export

c). `SELECT * FROM Students WHERE NOT department='ME';`

	student_id	name	email	phone	department
☐ Edit Copy Delete	1	Arun	arun@gmail.com	9876543210	CSE
☐ Edit Copy Delete	2	Bala	bala@gmail.com	9876543211	ISE
☐ Edit Copy Delete	3	Charan	charan@gmail.com	9876543212	ECE
☐ Edit Copy Delete	4	Divya	divya@gmail.com	9876543213	CSE
☐ Edit Copy Delete	5	Esha	esha@gmail.com	9876543214	ISE
☐ Edit Copy Delete	7	Gokul	gokul@gmail.com	9876543216	CSE
☐ Edit Copy Delete	8	Hari	hari@gmail.com	9876543217	ECE
☐ Edit Copy Delete	9	Isha	isha@gmail.com	9876543218	ISE
☐ Edit Copy Delete	10	John	john@gmail.com	9876543219	CSE

↑ ☐ Check all With selected: Edit Copy Delete Export

3 Aggregate functions: MIN, MAX, COUNT, AVG, SUM

a). count

Your SQL query has been executed successfully

`SELECT COUNT(*) FROM Students;`

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain](#)]

Extra options

COUNT(*)

10

b). MINIMUM

```
SELECT MIN(marks) FROM Marks;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain](#)]

☐ Show all | Number of rows:

Extra options

MIN(marks)
55

c). MAXIMUM

```
SELECT MAX(marks) FROM Marks;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain](#)]

☐ Show all | Number of rows:

Extra options

MAX(marks)
95

d). AVERAGE

```
SELECT AVG(marks) FROM Marks;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain](#)]

☐ Show all | Number of rows:

Extra options

AVG(marks)
76.2500

e). SUM

```
SELECT sum(marks) FROM Marks;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain](#)]

☐ Show all | Number of rows:

Extra options

sum(marks)
1525

4 Filtering: LIKE, IN, BETWEEN

a). SELECT * FROM Students WHERE name LIKE 'A%';

	student_id	name	email	phone	department
☐ Edit Copy Delete	1	Arun	arun@gmail.com	9876543210	CSE

↑ ☐ Check all With selected: Edit Copy Delete Export

b). SELECT * FROM Students WHERE department IN ('CSE','ISE');

	student_id	name	email	phone	department
☐ Edit Copy Delete	1	Arun	arun@gmail.com	9876543210	CSE
☐ Edit Copy Delete	2	Bala	bala@gmail.com	9876543211	ISE
☐ Edit Copy Delete	4	Divya	divya@gmail.com	9876543213	CSE
☐ Edit Copy Delete	5	Esha	esha@gmail.com	9876543214	ISE
☐ Edit Copy Delete	7	Gokul	gokul@gmail.com	9876543216	CSE
☐ Edit Copy Delete	9	Isha	isha@gmail.com	9876543218	ISE
☐ Edit Copy Delete	10	John	john@gmail.com	9876543219	CSE

↑ ☐ Check all With selected: Edit Copy Delete Export

c). SELECT * FROM Marks WHERE marks BETWEEN 70 AND 90;

	mark_id	student_id	course_id	marks	grade
☐ Edit Copy Delete	1	1	101	85	A
☐ Edit Copy Delete	2	1	102	78	B
☐ Edit Copy Delete	3	2	101	88	A
☐ Edit Copy Delete	5	4	104	90	A
☐ Edit Copy Delete	6	5	105	72	B
☐ Edit Copy Delete	10	9	104	80	B
☐ Edit Copy Delete	11	10	105	70	B
☐ Edit Copy Delete	13	3	104	75	B
☐ Edit Copy Delete	14	4	105	82	A
☐ Edit Copy Delete	15	5	101	77	B
☐ Edit Copy Delete	18	8	104	73	B
☐ Edit Copy Delete	20	10	101	85	A

↑ ☐ Check all With selected: Edit Copy Delete Export

5 Joins: INNER, LEFT, RIGHT, CROSS, SELF

a).

```
SELECT Students.name, Courses.course_name
FROM Enrollments
INNER JOIN Students ON Enrollments.student_id = Students.student_id
INNER JOIN Courses ON Enrollments.course_id = Courses.course_id;
```

name	course_name
Arun	DBMS
Bala	DBMS
Farhan	DBMS
Esha	DBMS
Arun	Java
Gokul	Java
Charan	Python
Hari	Python
Bala	Python
Divya	Maths
Isha	Maths
Charan	Maths
Esha	OS
John	OS
Divya	OS

b).

```
SELECT Students.name, Courses.course_name
FROM Students
LEFT JOIN Enrollments ON Students.student_id = Enrollments.student_id
LEFT JOIN Courses ON Enrollments.course_id = Courses.course_id;
```

name	course_name
Arun	DBMS
Arun	Java
Bala	DBMS
Bala	Python
Charan	Python
Charan	Maths
Divya	Maths
Divya	OS
Esha	OS
Esha	DBMS
Farhan	DBMS
Gokul	Java
Hari	Python
Isha	Maths
John	OS

c).

```
SELECT Students.name, Courses.course_name
FROM Students
RIGHT JOIN Enrollments ON Students.student_id = Enrollments.student_id
RIGHT JOIN Courses ON Enrollments.course_id = Courses.course_id;
```

name	course_name
Arun	DBMS
Bala	DBMS
Farhan	DBMS
Esha	DBMS
Arun	Java
Gokul	Java
Charan	Python
Hari	Python
Bala	Python
Divya	Maths
Isha	Maths
Charan	Maths
Esha	OS
John	OS
Divya	OS

6 Grouping: GROUP BY, HAVING

a).

```
SELECT student_id, AVG(marks)
FROM Marks
GROUP BY student_id;
```

	student_id	AVG(marks)
☐ Edit Copy Delete	1	81.5000
☐ Edit Copy Delete	2	76.5000
☐ Edit Copy Delete	3	71.0000
☐ Edit Copy Delete	4	86.0000
☐ Edit Copy Delete	5	74.5000
☐ Edit Copy Delete	6	64.5000
☐ Edit Copy Delete	7	93.0000
☐ Edit Copy Delete	8	64.0000
☐ Edit Copy Delete	9	74.0000
☐ Edit Copy Delete	10	77.5000

☐ Check all With selected: Edit Copy

b).

```
SELECT student_id, AVG(marks)
FROM Marks
GROUP BY student_id
HAVING AVG(marks) > 75;
```


← T →				student_id	AVG(marks)	
☐		Edit	Copy	Delete	1	81.5000
☐		Edit	Copy	Delete	2	76.5000
☐		Edit	Copy	Delete	4	86.0000
☐		Edit	Copy	Delete	7	93.0000
☐		Edit	Copy	Delete	10	77.5000

7 Advanced: UNION, EXISTS, CASE

a).

```
SELECT name FROM Students
UNION
SELECT course_name FROM Courses;
```

name
Arun
Bala
Charan
Divya
Esha
Farhan
Gokul
Hari
Isha
John
DBMS
Java
Python
Maths
OS

b) .

```
SELECT name
FROM Students
WHERE EXISTS (
SELECT * FROM Marks
WHERE Marks.student_id = Students.student_id
);
```

	name
☐  Edit  Copy  Delete	Arun
☐  Edit  Copy  Delete	Bala
☐  Edit  Copy  Delete	Charan
☐  Edit  Copy  Delete	Divya
☐  Edit  Copy  Delete	Esha
☐  Edit  Copy  Delete	Farhan
☐  Edit  Copy  Delete	Gokul
☐  Edit  Copy  Delete	Hari
☐  Edit  Copy  Delete	Isha
☐  Edit  Copy  Delete	John

c).

```

SELECT student_id, marks,
CASE
WHEN marks >= 90 THEN 'A'
WHEN marks >= 75 THEN 'B'
ELSE 'C'
END AS Grade
FROM Marks;

```

	student_id	marks	Grade
☐  Edit  Copy  Delete	1	85	B
☐  Edit  Copy  Delete	1	78	B
☐  Edit  Copy  Delete	2	88	B
☐  Edit  Copy  Delete	3	67	C
☐  Edit  Copy  Delete	4	90	A
☐  Edit  Copy  Delete	5	72	C
☐  Edit  Copy  Delete	6	60	C
☐  Edit  Copy  Delete	7	95	A
☐  Edit  Copy  Delete	8	55	C
☐  Edit  Copy  Delete	9	80	B
☐  Edit  Copy  Delete	10	70	C
☐  Edit  Copy  Delete	2	65	C
☐  Edit  Copy  Delete	3	75	B
☐  Edit  Copy  Delete	4	82	B
☐  Edit  Copy  Delete	5	77	B
☐  Edit  Copy  Delete	6	69	C
☐  Edit  Copy  Delete	7	91	A
☐  Edit  Copy  Delete	8	73	C

Part 4: Data Modification

Perform UPDATE, DELETE, and INSERT SELECT operations.

a). UPDATE Students SET department='CSE' WHERE student_id=1;

Before :

	student_id	name	email	phone	department
☐ Edit Copy Delete	1	Arun	arun@gmail.com	9876543210	ISE

After :

	student_id	name	email	phone	department
☐ Edit Copy Delete	1	Arun	arun@gmail.com	9876543210	CSE

b). DELETE FROM Students WHERE student_id=10;

Before :

	student_id	name	email	phone	department
☐ Edit Copy Delete	1	Arun	arun@gmail.com	9876543210	CSE
☐ Edit Copy Delete	2	Bala	bala@gmail.com	9876543211	ISE
☐ Edit Copy Delete	3	Charan	charan@gmail.com	9876543212	ECE
☐ Edit Copy Delete	4	Divya	divya@gmail.com	9876543213	CSE
☐ Edit Copy Delete	5	Esha	esha@gmail.com	9876543214	ISE
☐ Edit Copy Delete	6	Farhan	farhan@gmail.com	9876543215	ME
☐ Edit Copy Delete	7	Gokul	gokul@gmail.com	9876543216	CSE
☐ Edit Copy Delete	8	Hari	hari@gmail.com	9876543217	ECE
☐ Edit Copy Delete	9	Isha	isha@gmail.com	9876543218	ISE
☐ Edit Copy Delete	10	John	john@gmail.com	9876543219	CSE

After :

	student_id	name	email	phone	department
☐ Edit Copy Delete	1	Arun	arun@gmail.com	9876543210	CSE
☐ Edit Copy Delete	2	Bala	bala@gmail.com	9876543211	ISE
☐ Edit Copy Delete	3	Charan	charan@gmail.com	9876543212	ECE
☐ Edit Copy Delete	4	Divya	divya@gmail.com	9876543213	CSE
☐ Edit Copy Delete	5	Esha	esha@gmail.com	9876543214	ISE
☐ Edit Copy Delete	6	Farhan	farhan@gmail.com	9876543215	ME
☐ Edit Copy Delete	7	Gokul	gokul@gmail.com	9876543216	CSE
☐ Edit Copy Delete	8	Hari	hari@gmail.com	9876543217	ECE
☐ Edit Copy Delete	9	Isha	isha@gmail.com	9876543218	ISE
☐ Check all	With selected: Edit Copy Delete Export				

c).

```
INSERT INTO Marks (student_id, course_id, marks, grade)
```

```
SELECT student_id, course_id, 0, 'C'
```

```
FROM Enrollments;
```

Before :

				enrollment_id	student_id	course_id
☐		Edit		Copy		Delete
				1	1	101
☐		Edit		Copy		Delete
				2	1	102
☐		Edit		Copy		Delete
				3	2	101
☐		Edit		Copy		Delete
				4	3	103
☐		Edit		Copy		Delete
				5	4	104
☐		Edit		Copy		Delete
				6	5	105
☐		Edit		Copy		Delete
				7	6	101
☐		Edit		Copy		Delete
				8	7	102
☐		Edit		Copy		Delete
				9	8	103
☐		Edit		Copy		Delete
				10	9	104
☐		Edit		Copy		Delete
				12	2	103
☐		Edit		Copy		Delete
				13	3	104
☐		Edit		Copy		Delete
				14	4	105
☐		Edit		Copy		Delete
				15	5	101

14/16 selected

After :

				enrollment_id	student_id	course_id
☐		Edit		Copy		Delete
				1	1	101
☐		Edit		Copy		Delete
				2	1	102
☐		Edit		Copy		Delete
				3	2	101
☐		Edit		Copy		Delete
				4	3	103
☐		Edit		Copy		Delete
				5	4	104
☐		Edit		Copy		Delete
				6	5	105
☐		Edit		Copy		Delete
				7	6	101
☐		Edit		Copy		Delete
				8	7	102
☐		Edit		Copy		Delete
				9	8	103
☐		Edit		Copy		Delete
				10	9	104
☐		Edit		Copy		Delete
				12	2	103
☐		Edit		Copy		Delete
				13	3	104
☐		Edit		Copy		Delete
				14	4	105
☐		Edit		Copy		Delete
				15	5	101

Part 5: Advanced Features

Views

```
CREATE VIEW student_view AS
SELECT name, department FROM Students;
SELECT * FROM student_view;
```

←T→ ▼ name department			
☐	 Edit	 Copy	 Delete
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☐	 Edit	 Copy	 Delete
☐	 Edit	 Copy	 Delete
☐	 Edit	 Copy	 Delete
☐	 Edit	 Copy	 Delete
☐	 Edit	 Copy	 Delete
☐	 Edit	 Copy	 Delete
☐	 Edit	 Copy	 Delete
☐	 Edit	 Copy	 Delete
⬆ ☐ Check all With selected:  Edit  Copy			